22.8. NATURAL ANALOGUES

The geochemical modeling calculations based on measured equilibrium data are of primary importance in the safety assessments of proposed nuclear waste repositories (and should be important for other waste repositories as well). Another useful tool in such assessments is the data from studies of natural analogue sites.

The modeling calculations use data from laboratory and field studies which have been obtained over the last few decades. These data are used in the codes to predict the solubilities, and nuclide migration of material which might be released from nuclear repositories over thousands and, even, hundreds of thousands of years. It is not possible to demonstrate rigorously that the models used are accurate as they may simplify the natural system, use incorrect data or misrepresent (possibly, ignore) important processes which occur over very long time periods. However, some validation of the models and data used in the modeling calculations can be obtained from careful comparison of calculated values with those measured at appropriate geologic sites, known as a natural analogues.

The natural analogue sites are areas in which uranium ores have been present for geologic time periods. In most cases, these sites have not been affected by human activities, so the record of geologic, long term effects are well preserved. A number of such sites are being studied around the world; studies from a few sites of different characteristics are reviewed here.

From studies of granitic sites, the dominant role of fractures and fissures in the transport of fluids has been convincingly demonstrated. Thus any model of water in a granitic site must include both advection which is dominant in the fractures and fissures, and diffusion which is important in regions of highly altered rocks ("alteration rims"). In clays, mass transport seems to proceed primarily by ionic and molecular diffusion although some fluid mass transport occurs at discontinuities in the formation.

Often, the mobilization of many elements (e.g. U, but not Th, in crystalline rocks) is correlated with the flow of oxidizing water. The mobilization and fixation of uranium involves complexation, redox and retention on minerals via adsorption, and ion exchange. In clay media, the redox potential is strongly buffered if significant amounts of organic substances are present.

In a section of the Pocos de Caldas (Brazil) formation, most of the thorium and the rare earths, and, to a lesser extent, the uranium, is associated with goethite (FeOOH) particles and transport by organic colloids is much less important. This region is reducing, as shown by the presence of Fe(II). However, in another region of this formation, the thorium and rare earths have a much higher mobility and are associated with organic (humic) colloids. At an analogue site in Scotland, the flow from the ore has passed into a peat bog in which the uranium is associated predominantly with humic material while the Th is found on Fe/Al oxyhydroxide colloids and particles. In many

clay deposits, the organic material is most significant in maintaining a reducing potential which restricts actinide migration and provides a sorption source of the mobilized fraction.

22.9. THE OKLO REACTOR

Analysis of the Oklo natural reactors (§19.10) indicates that they must have lasted for 100 000 to 500 000 years with criticality occurring periodically. They probably would have consumed about twelve metric tons of ^{235}U, releasing a total energy of 2–3 GWy at a probable power level of ~10 kW. About 1.0–1.5 t of ^{239}Pu were formed by neutron capture in ^{238}U, but the half-life of ^{239}Pu has resulted in its total decay to ^{235}U. Since a few samples enriched in ^{235}U (presumably due to this decay) have been found, it is believed that in some places in Oklo breeding conditions may have temporarily existed. The decaying fission products are estimated to have had 10^{28} β,γ disintegrations over the operating time of the reactors. The average energy release in the reactor zone was ~50 W m^{-2} which is several times greater than that planned for geological nuclear waste repositories. As a consequence, it is estimated that the fluids in the inclusions of the mineral grains in the Oklo reactors had temperatures of 450 to 600°C, well above those anticipated in deep waste repositories. There is evidence of connectively driven circulation of the fluids for distances of 30 m from the main reactor zones as well as significant dissolution and modification of minerals due to the thermal and radiation conditions. Redistribution of some elements resulted from the convective flow of the hot liquids.

The uranium minerals appear to have remained relatively stable despite this heating process. The uranium and the lanthanide elements show evidence of some small degree of localized redistribution, but were mostly retained within the reactor zone. By contrast, fission product rare gases, halogens, molybdenum and the alkali and alkaline earth elements migrated significant distances from the reactor zones. In general, it seems that elemental redistribution took place over a period of 0.5 to million years while the area was thermally hot (during and after nuclear criticality). It has been estimated that as much as a total of 10^{12} liter of hot water flowed through the reactor zones. The water leaving the reactor zones had 5×10^{-3} g U m^{-3} and ~10^{-10} M concentrations of Tc, Ru, and Nd. The Tc and Ru were oxidized to TcO_4^- and RuO_4^-, and these soluble oxyanions moved with the water as it flowed from the area, creating significant (25–35%) deficiencies of these elements in the reactor zones. The Nd was less soluble and, apparently, migrated much less during this hot period. For Tc and Ru, the migration rate seems to have been ~10^{-5} m y^{-1} in water moving at a flow of 5 m y^{-1}.

A very important observation is the evidence in Oklo that the plutonium produced by the reactors did not move during its lifetime from the site of its formation.